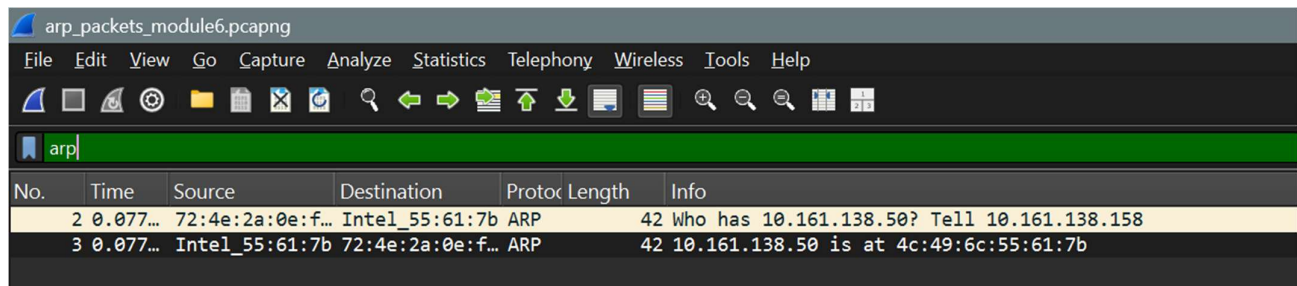


Structured Training Program: Networking**Week 3***Module 6: Layer 3 Session 2*

1. Capture and analyze ARP packets in Wireshark. Inspect the ARP request and reply frames when your device attempts to find the router's MAC address.

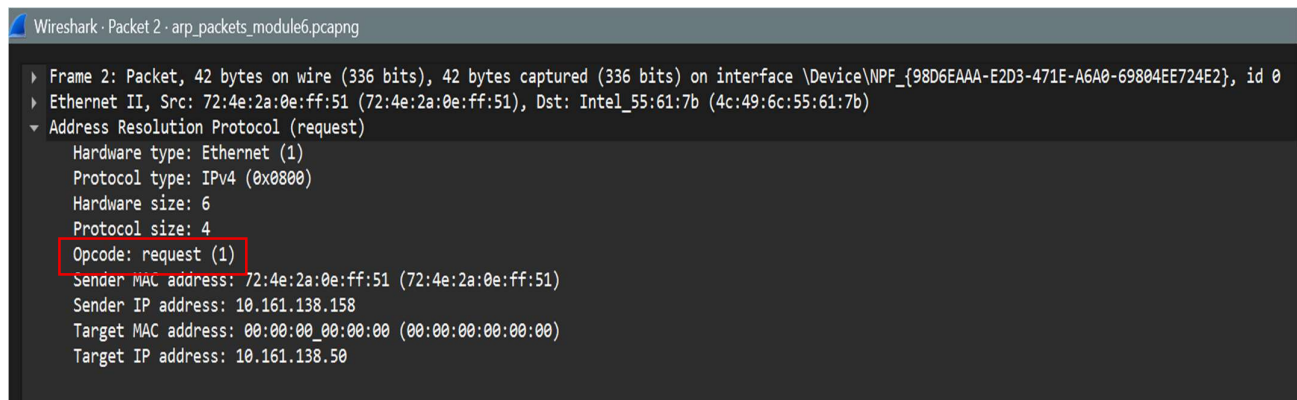
Discuss the importance of ARP in packet forwarding.

I have captured the following ARP packets



No.	Time	Source	Destination	Protocol	Length	Info
2	0.077...	72:4e:2a:0e:f...	Intel_55:61:7b	ARP	42	Who has 10.161.138.50? Tell 10.161.138.158
3	0.077...	Intel_55:61:7b	72:4e:2a:0e:f...	ARP	42	10.161.138.50 is at 4c:49:6c:55:61:7b

Inspecting the request ARP packet,



```

Wireshark · Packet 2 · arp_packets_module6.pcapng
  ▶ Frame 2: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{98D6EAAA-E2D3-471E-A6A0-69804EE724E2}, id 0
  ▶ Ethernet II, Src: 72:4e:2a:0e:ff:51 (72:4e:2a:0e:ff:51), Dst: Intel_55:61:7b (4c:49:6c:55:61:7b)
  ▼ Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: 72:4e:2a:0e:ff:51 (72:4e:2a:0e:ff:51)
    Sender IP address: 10.161.138.158
    Target MAC address: 00:00:00 00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.161.138.50
  
```

The packet consists of the following fields

Hardware Type (2 bytes): Specifies the type of data link layer technology used. It allows ARP to operate across different network technologies. For Ethernet networks, this value is 1.

Protocol Type (2 bytes): Identifies the network protocol whose address is being mapped to a MAC address. It is commonly IPv4 (0x0800). **Hardware Size (1 byte):** Specifies the length of the hardware address. For Ethernet, the MAC address length is 6 bytes (48 bits).

Protocol Size (1 byte): Specifies the length of the protocol address. For IPv4, the IP address length is 4 bytes (32 bits).

Opcode (2 bytes): Indicates the type of ARP message.

- 1 – ARP Request
- 2 – ARP Reply

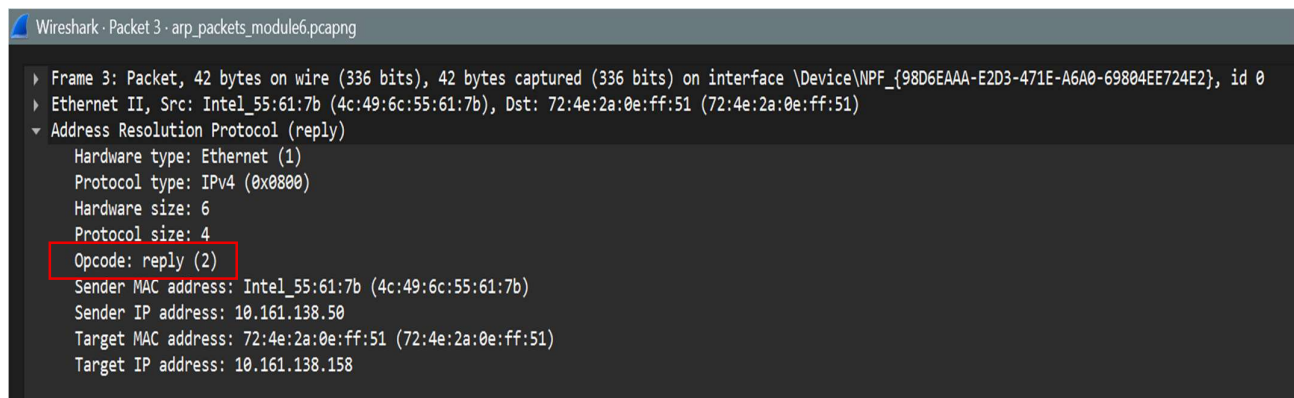
Sender MAC Address (6 bytes): Specifies the MAC address of the sender (the device that generated the ARP request).

Sender IP Address (4 bytes): Specifies the IP address of the sender. This allows the receiving device to know where to send the ARP reply and helps the receiver device to update its ARP cache if needed.

Target MAC Address (6 bytes): In an ARP request, this field is typically set to 00:00:00:00:00:00 because the sender does not yet know the MAC address of the target device.

Target IP Address (4 bytes): Specifies the IP address of the device whose MAC address is being requested. The ARP request is broadcast to all devices in the local network to find this address.

Inspecting the ARP reply packet,



```
Wireshark - Packet 3 - arp_packets_module6.pcapng
> Frame 3: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{98D6EAAA-E2D3-471E-A6A0-69804EE724E2}, id 0
> Ethernet II, Src: Intel_55:61:7b (4c:49:6c:55:61:7b), Dst: 72:4e:2a:0e:ff:51 (72:4e:2a:0e:ff:51)
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Intel_55:61:7b (4c:49:6c:55:61:7b)
  Sender IP address: 10.161.138.50
  Target MAC address: 72:4e:2a:0e:ff:51 (72:4e:2a:0e:ff:51)
  Target IP address: 10.161.138.158
```

The packet structure of the **ARP reply** is similar to that of the **ARP request**. However, the **main difference** is that in the reply packet, the **Target MAC Address field contains the actual MAC address of the requesting system**, instead of being set to 00:00:00:00:00:00 as in the request packet.

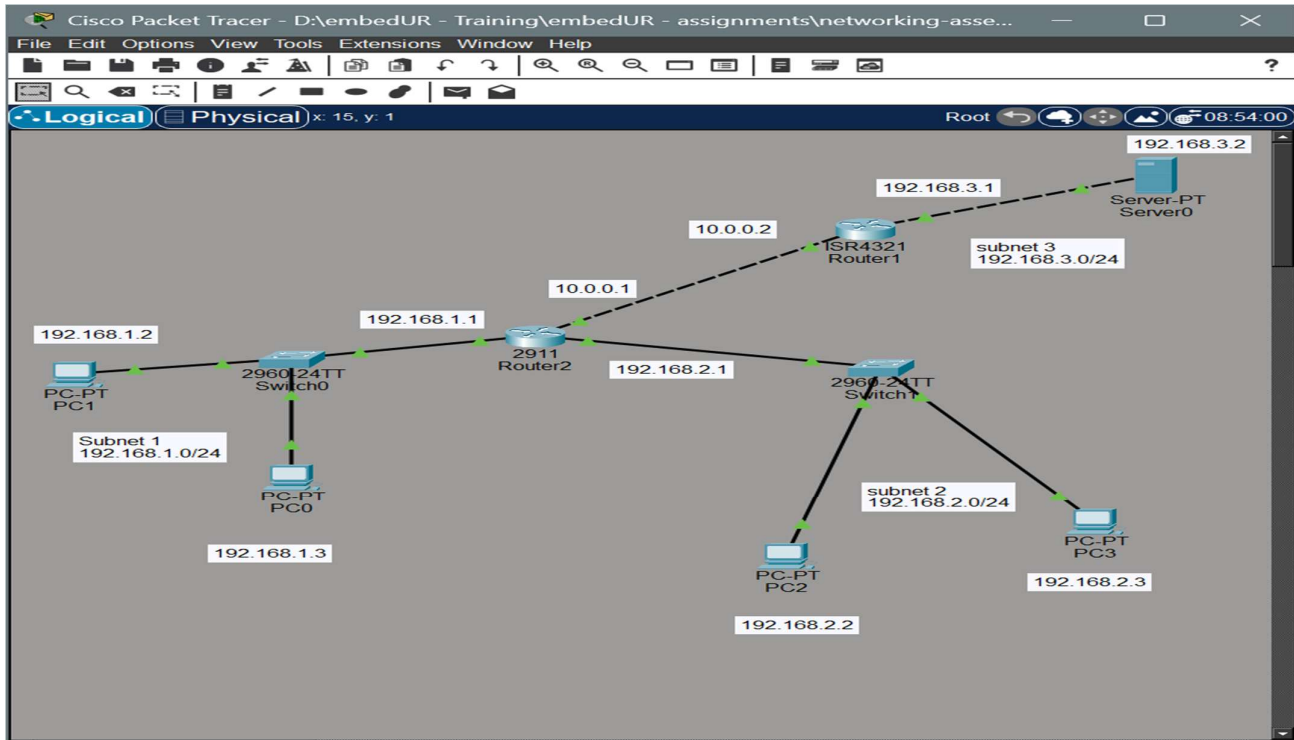
Additionally, the **Sender MAC Address field contains the MAC address of the device whose IP address was requested**, indicating the mapping between the IP address and its corresponding MAC address. The **ARP reply** is then sent as a **unicast packet** directly to the requesting system.

Importance of ARP in packet Forwarding

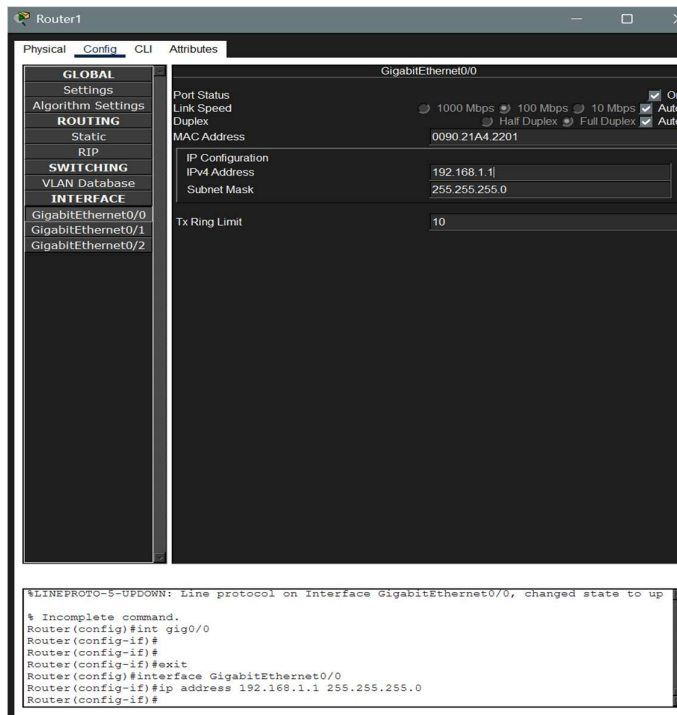
1. Maps Logical to Physical Addresses
2. Enables Local Network Delivery (as within Local Network the switch needs MAC for delivery)

2. Manually configure static routes on a router to direct packet to different subnets. Use the ip route command and verify connectivity using ping and traceroute

a) I have used the following topology to configure static routing



b) Now configure the IP and gateway in each PC and router as mentioned in the topology



c) Now add static routes in each router using the following commands,

The image shows two side-by-side screenshots of Cisco Packet Tracer router configuration windows. The left window is for Router2 and the right window is for Router1. Both windows show the configuration of interfaces and the addition of static routes. In Router2, the static route for 192.168.3.0/24 is highlighted with a red box. In Router1, the static routes for 192.168.1.0/24 and 192.168.2.0/24 are highlighted with a red box.

```

Router2
Router>enable
Router#
Router#configure terminal
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
ip address 10.0.0.2 255.0.0.0
Router(config-if)#ip address 10.0.0.2 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#exit
Router(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.1
Router(config)#

Router1
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#
Router(config)#interface GigabitEthernet0/2
Router(config-if)#shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
ip address 10.0.0.1 255.255.255.0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
Router(config-if)#

```

d) Now ping the machine in subnet 192.138.3.0/24

The image shows a screenshot of a PC1 Command Prompt window. The window title is "PC1" and it has tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt shows the results of a ping command to 192.168.3.2 and a traceroute command to 192.168.3.2. The ping command shows a 25% loss (1 out of 4 packets received). The traceroute command shows the path from 192.168.1.1 to 10.0.0.2 to 192.168.3.2.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.2: bytes=32 time=11ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>tracert 192.168.3.2

Tracing route to 192.168.3.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.1.1
  1  0 ms    0 ms    0 ms    10.0.0.2
  2  0 ms    0 ms    0 ms    192.168.3.2

Trace complete.

C:\>

```

3. Given a network address of 10.0.0.0/24, divide it into 4 equal subnets

- Calculate the new subnet mask.
- Determine the valid host range.
- Assign IP address to devices in Packet Tracer and verify connectivity.

i. Calculate the new Subnet mask:

Current Subnet Mask: Decimal: 255.255.255.0; Binary: 11111111.11111111.11111111.00000000

The no of subnet bits 'n' must satisfy, $2^n \geq$ no. of required subnets

So we get $n \Rightarrow 2$

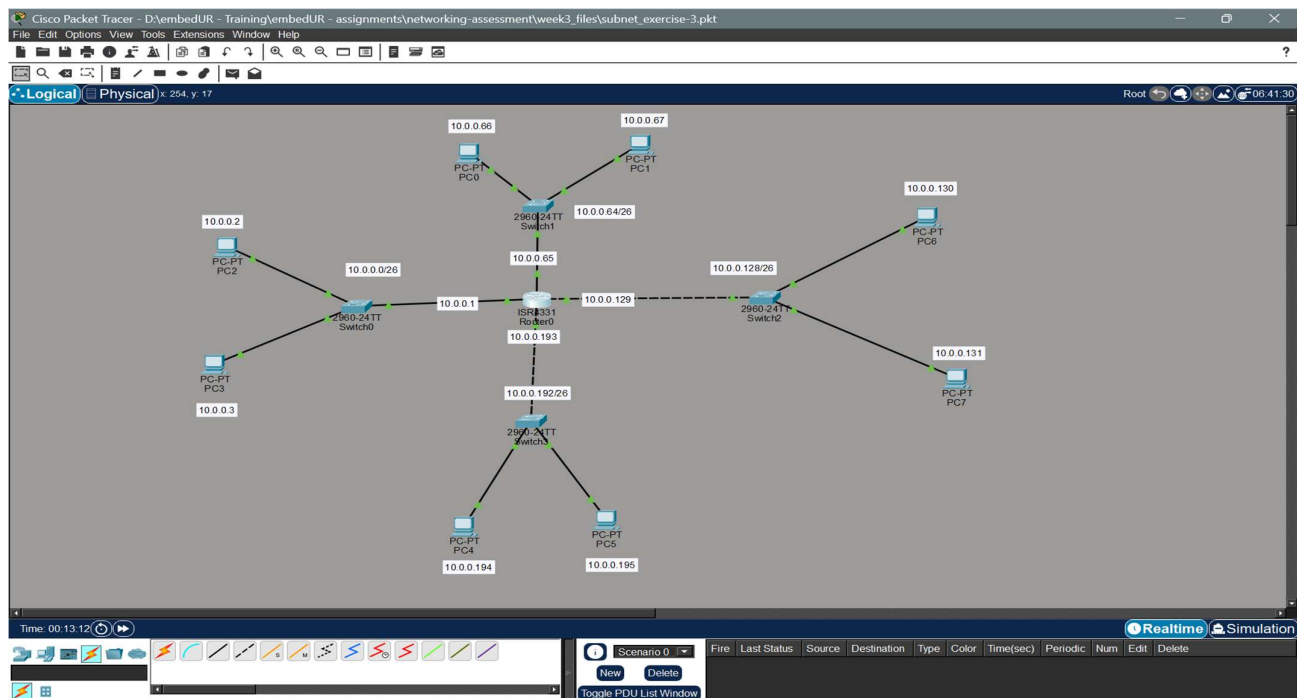
New Subnet Mask: Decimal: 255.255.255.192; Binary: 11111111.11111111.11111111.11000000

ii. Determine the valid host range

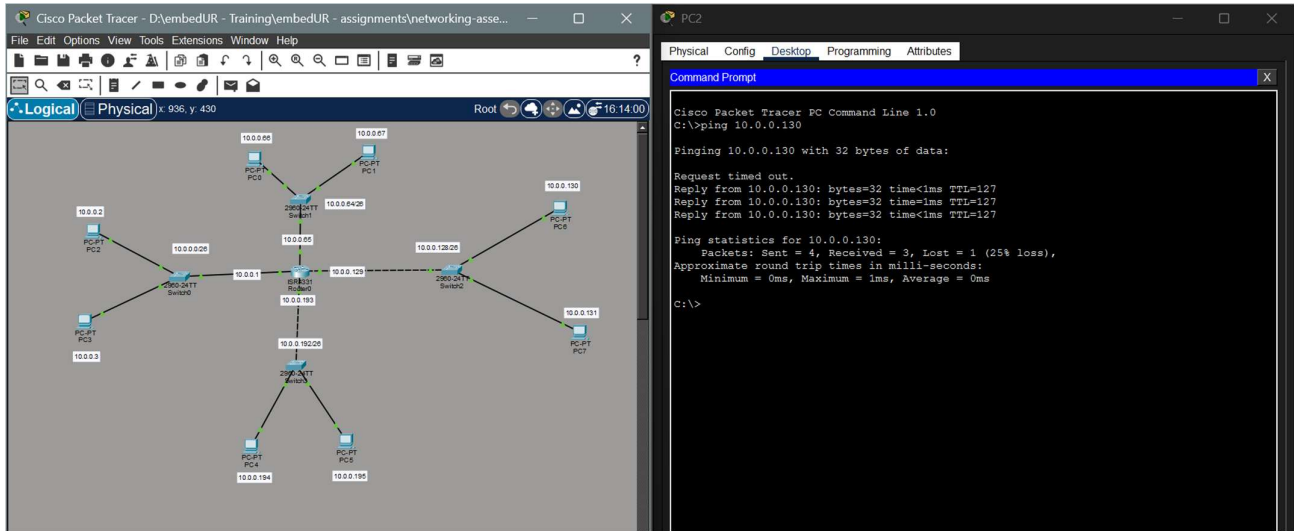
Subnet	Network ID	First Usable Host	Last Usable Host	Broadcast Address
1	10.0.0.0	10.0.0.1	10.0.0.62	10.0.0.63
2	10.0.0.64	10.0.0.65	10.0.0.126	10.0.0.127
3	10.0.0.128	10.0.0.129	10.0.0.190	10.0.0.191
4	10.0.0.192	10.0.0.193	10.0.0.254	10.0.0.255

iii. Assign IP address to devices in packet tracer and verify the connectivity

I have used the following topology and assigned PC's IP and gateway accordingly.



Checking the connectivity



4. You are given three IP addresses: 192.168.10.5, 172.20.15.1 and 8.8.8.8

Identify the class of each IP address

Determine if it is a private or public IP

Explain how NAT would handle a private IP when accessing the internet

The classes of the given IP address and its types are as follow

IP Address	Class	Type	Reason
192.168.10.5	Class C	Private	Falls within the Class C private range (192.168.0.0 – 192.168.255.255).
172.20.15.1	Class B	Private	Falls within the Class B private range (172.16.0.0 – 172.31.255.255).
8.8.8.8	Class A	Public	Falls outside the Class A private range (10.x.x.x) and it is a well known Google DNS IP

How NAT Handles Private IPs

Network Address Translation (NAT) acts as a mediator between a local private network and the public internet. Since private IP addresses are not "routable" on the public internet (meaning internet routers will drop packets coming from them), NAT is required for communication.

Steps Involved:

- Request:** When your device (e.g., 192.168.XXX.XXX) sends a request to a website, the packet reaches your router.
- Translation:** The router replaces your Private Source IP with its own Public IP address (assigned by your ISP).

3. **Mapping:** The router makes a note in its NAT Table, recording which internal device made the request and which port was used.
4. **Response:** When the website sends data back, it sends it to the router's Public IP. The router looks at its table, sees the request belongs to 192.168.10.5, translates the destination back to your private IP, and forwards the data to you.

Importance:

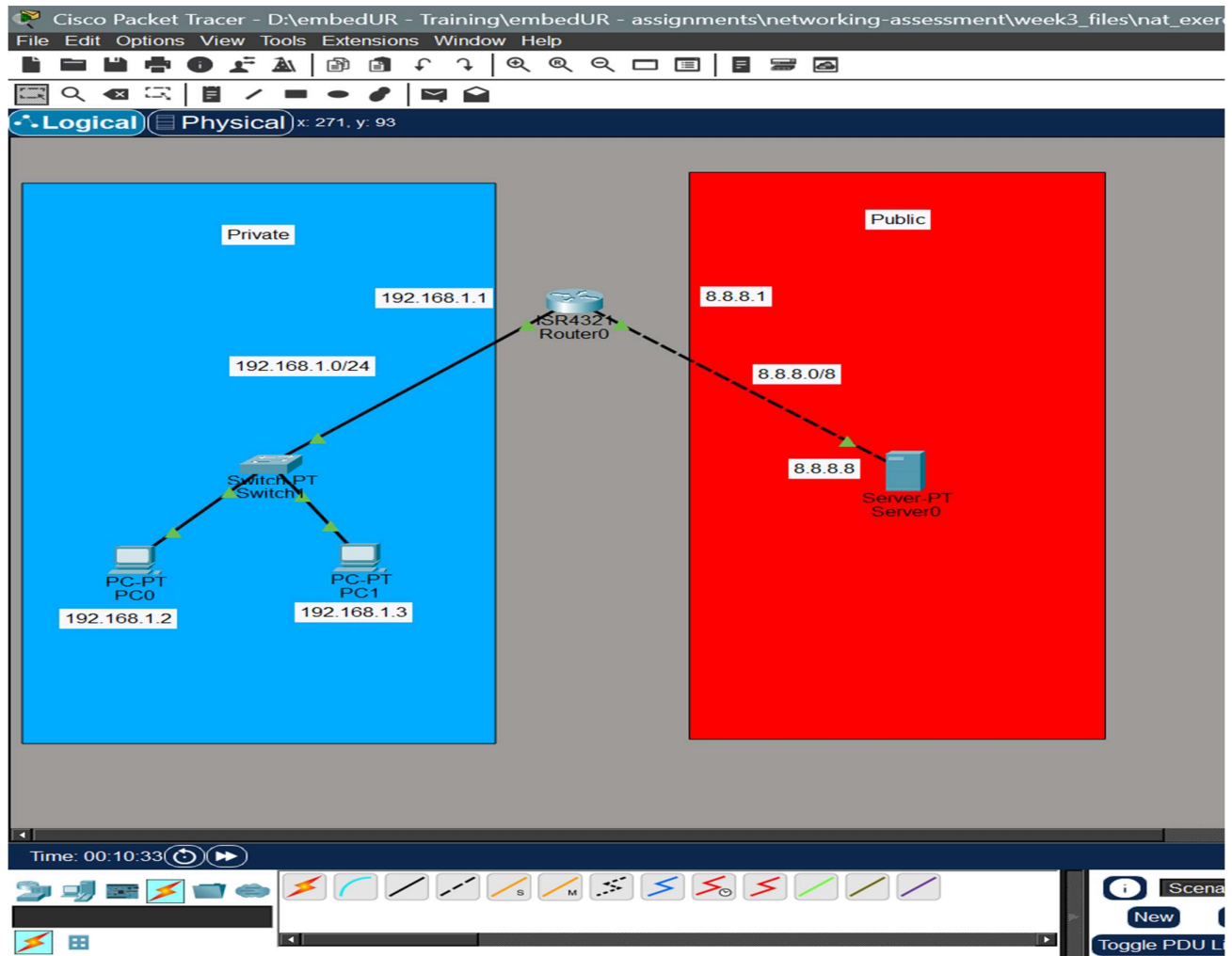
- It allows hundreds of devices to share a single public IP address (IP conservation).
- It hides the internal IP addresses of your network from the outside world (Security).

5. In CISCO Packet Tracer, configure NAT on a router to allow internal devices (192.168.1.x) to access the internet.

Test the connectivity by pinging an external public IP

Capture the traffic in wireshark and analyze the source IP before and after NAT translation.

To demonstrate the above NAT experiment I have used the following topology.



To create a NAT for a set of devices we need to

- Create a Access Control List of IPs allowed
- Create a pool of public IP available for translation
- Bind the IP address (in ACL) and the Pool of public addresses

```

Router0
Physical Config CLI Attributes
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:..

--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#acl-
Router(config-if)#acl-
Router(config-if)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)#ip nat pool NAT_POOL 20.0.0.1 20.0.0.10 netmask 255.255.255.0
Router(config)#ip nat inside source list 1 pool NAT_POOL
Router(config)#int g
Router(config)#int gigabitEthernet
Router(config)#int gigabitEthernet
Router(config)#int gig0/0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#

```

On Simulation mode I get the following packets at each stage

Simulation Panel

Vis.	Time(sec)	Last Device
0.000	-	
0.001	0.001	PC0
0.002	0.002	Switch1
0.003	0.003	Router0
0.004	0.004	Server0
0.005	0.005	Router0
0.006	0.006	Switch1

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, L2L3, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAM, POP3, PPP, PPPoE, PTP, Protocol, RADIUS, REP, RIPv2, RIPv3, RIPv6, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Simulation

Scenario 0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	Server0	ICMP		0.000	N	0	(edit)	(delete)

Let us see the packet in the LAN (private) and WAN (public) network.

The screenshot displays the Cisco Packet Tracer interface. On the left, a red area represents the 'Public' network, and a blue area represents the 'Private' network. A server labeled 'Server-PT Server0' is connected to the Public network. The 'PDU Information at Device: Server0' window is open, showing the 'Inbound PDU Details' tab. The details are as follows:

Layer	Details
Layer7	
Layer6	
Layer5	
Layer4	
Layer3	IP Header Src. IP: 20.0.0.1, Dest. IP: 8.8.8.8 ICMP Message Type: 8
Layer2	Ethernet II Header 0030.A3E7.6102 >> 0001.6476.30A5
Layer1	Port(s): FastEthernet0

The 'Simulation Panel' on the right shows the 'Event List' with the following data:

Vis.	Time(sec)	Last Device	At Device
	0.000	--	PC0
	0.001	PC0	Switch1
	0.002	Switch1	Router0
	0.003	Router0	Server0
	0.004	Server0	Router0
	0.005	Router0	Switch1
	0.006	Switch1	PC0
	1.998	--	Switch1

The 'Simulation Panel' also includes a 'Reset Simulation' button, a 'Constant Delay' checkbox, and a 'Captured to: 1.998 s' indicator. The 'Event List Filters - Visible Events' section lists various protocols and services supported by the simulation.

We can see that I have actually pinged from PC0 whose IP is 192.168.1.1 but as we have created a NAT Gateway, the IP sent to the server is 20.0.0.1. We can also see the NAT Table in the router for further confirmation.

The screenshot shows the Cisco Packet Tracer interface with the 'NAT Table for Router0' window open. The table displays the following information:

Protocol	Inside Global	Inside Local	Outside Local	Outside Global
icmp	20.0.0.1:11	192.168.1.2:11	8.8.8.8:11	8.8.8.8:11
icmp	20.0.0.1:2	192.168.1.2:2	8.8.8.8:2	8.8.8.8:2

The background shows the network topology with a 'Private' network (blue) and a 'Public' network (red). A switch is connected to the Private network, and a router is connected to both the Private and Public networks. The router's configuration shows the NAT Table for Router0.

We can also find that the server can never access the PC's in private network with their private IP.

The screenshot displays the Cisco Packet Tracer interface. On the left, a 'Command Prompt' window for 'Server0' is open, showing the following commands and output:

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.1
Invalid Command.

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Ping statistics for 192.168.1.1:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Ping statistics for 192.168.1.1:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),

Control-C
^C
C:\>
```

On the right, the 'Simulation Panel' is visible, showing an 'Event List' table. The table has four columns: 'Vis.', 'Time(sec)', 'Last Device', and 'At Device'. The events listed are:

Vis.	Time(sec)	Last Device	At Device
	0.000	--	PC0
	0.001	PC0	Switch1
	0.002	Switch1	Router0
	0.003	Router0	Server0
	0.004	Server0	Router0
	0.005	Router0	Switch1
	0.006	Switch1	PC0
	1.998	--	Switch1
	1.998	--	Server0
	1.998	--	Server0